Title: Ingestion of Workday API data in Armanino

Client name: Armanino

Industry: Industrial Software

Jade Roles & Responsibilities :

Architected & Implemented a Unified Ingestion Framework

- Designed a scalable, configuration-driven API ingestion framework in Databricks to extract data across disparate source systems (**Workday, Adaptive, LCVista, Confirmations, and CRM**) into a single, unified data repository.
- Handled complex, nested semi-structured JSON payloads natively in PySpark, flattening high-volume schema structures into optimized columnar formats for high-speed downstream processing.

Built High-Performance API Chunking & Incremental Ingestion Logic

- Engineered a metadata-driven chunking engine using centralized control tables to break large historical extractions (**Billing, Resource Forecast, Timeblocks, Invoices**) into configurable date/volume windows, preventing API throttling and rate-limit failures.
- Implemented incremental pipeline tracking (`incr version`) mapped via configuration tables to poll APIs delta-by-delta, enabling high-frequency, near-real-time data refreshes.

Optimized Data Pipeline Architecture & Reporting Performance

- Structured a **Staging-to-Target (Gold/Data Mart) architecture** where continuous background extraction writes to staging, executing one-shot atomic updates into target tables to isolate active reporting tables from locked writes.
- Tuned Spark business logic and automated pipeline triggers to drastically reduce Power BI dataset refresh times to **under 1 minute**, meeting strict end-user SLAs for near-real-time reporting.

End-to-End Delivery & Power BI Data Mart Modeling

- Led the full project lifecycle — from technical architecture, design, and PySpark development to automated pipeline orchestration and Power BI modeling.

- Modeled enterprise data marts combining Workday, CRM, and financial feeds, providing unified cross-functional visibility without cross-source query friction.

Key Technologies/Platform:ADF,Databricks

At a Glance (summary): Create Configurable Code which will use Databricks as compute and ADF pipelines as Orchestrator to ingest Workday Object in SQL server.

About Client : Armanino is an independent accounting and business consulting firm serving privately-held companies, as well as non-profit organizations and public entities with assurance/audit, tax, consulting and business management services.

It is one of the 20 largest independent accounting and business consulting firms in the United States which was founded in 1969, the firm generates over $700 million in annual revenue and operates globally, including expanding office locations in India.

Business Challenges:

1. **Scalability Bottlenecks and High Latency with Traditional ETL Engines**
   Traditional ETL tools like Azure Data Factory (ADF) struggled to efficiently handle massive data volumes and complex processing, resulting in long execution times, frequent resource constraints, and operational delays.
2. **Difficulty Ingesting and Parsing Complex Nested API Structures**
   Incoming data payloads from Workday APIs contained highly complex nested structures and array objects. Traditional ETL engines failed to parse these semi-structured formats cleanly, leading to high processing overhead and potential data loss.
3. **Inability to Support Near-Real-Time Data Availability Requirements**
   Legacy batch pipelines relied heavily on full target reloads, which took too long to run. This created a delay in data availability, preventing downstream business users and analytics platforms from accessing current insights.
4. **Downtime and Query Disruptions During Frequent Target Data Refreshes**
   Updating target tables directly during data refresh cycles risk locking tables, causing query timeouts, and interrupting active business users who need continuous access to master records.
5. **API Timeout Errors and Data Loss Caused by Large Volume Ingestion**
   Attempting to retrieve large datasets in single API calls from Workday frequently triggered system timeouts. This caused pipeline failures, missing data chunks, and high operational effort to manually restart failed loads.

Business Requirements:

1. Near Real Time Data Integration and Synchronization

Integrate Workday API near real time and users should have a seamless experience with latest data updates.

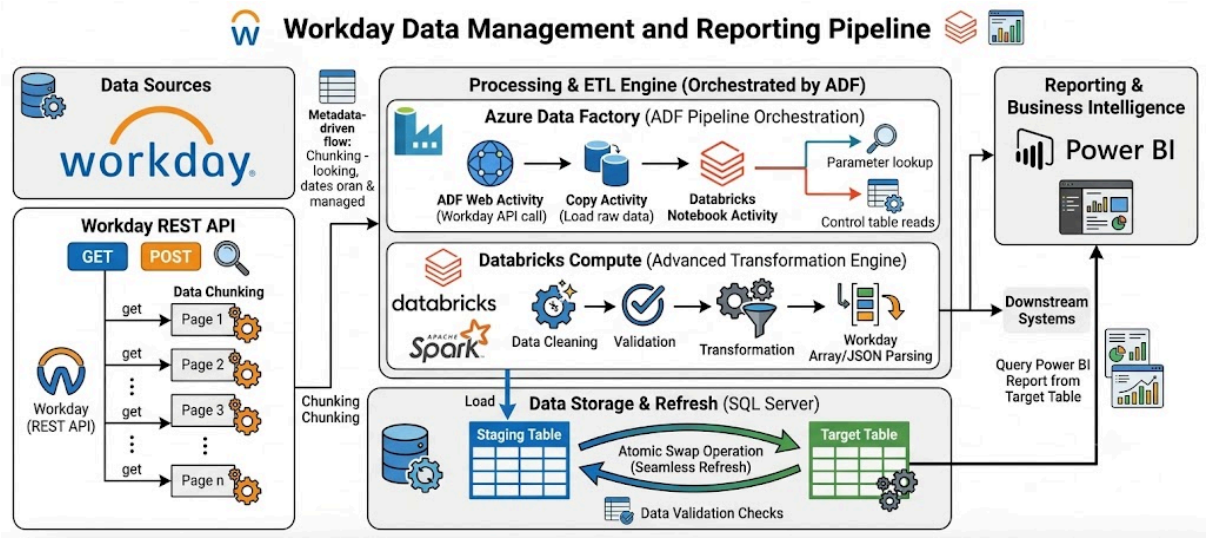
2. Data Quality Assurance with no data loss

Not a single row should be missed -this was frequent given API timeout's due to huge data.Complex objects should also not get missed as some columns can be inside lists/arrays.

3. Accelerated Development with minimum maintenance:

Creating a framework was necessary to minimize maintenance given similar code structures for various databricks objects

Our Approach and solutions :



1. **Scalable Databricks Compute Engine**
   We implemented **Databricks as a high-performance compute platform** to process large-scale datasets concurrently. Leveraging Apache Spark's distributed architecture, Databricks handled heavy transformation workloads

with ease, drastically cutting execution times and overcoming ADF's compute constraints.

2. **Native Complex Data Type Handling**
   We utilized Databricks' robust native support for **complex data structures**. This enabled us to efficiently explode, flatten, and parse nested arrays and multi-level JSON payloads directly within Spark DataFrames without performance degradation or complex custom code.
3. **Incremental Data Pipelines for Near-Real-Time Delivery**
   We engineered **incremental processing pipelines using Databricks compute**. By capturing and processing only changed or newly modified records since the last run, we cut down pipeline runtime significantly and delivered near-real-time updates to downstream systems.
4. **Seamless Target Refreshes via Atomic Table Swaps**
   We designed a **zero-downtime refresh architecture using temporary staging tables**. Databricks loads transformed data into a staging table first; once complete, the data is atomically swapped with the primary target table. This guarantees zero query interruptions and a seamless experience for end consumers.
5. **Data Chunking Strategy via Metadata-Driven Control Table**
   We built a **metadata-driven framework using a single, centralized control table** that holds API parameters, windowing start/end dates, and chunk size definitions. The pipeline ingests data in smaller, controlled time slices (chunking), eliminating API timeouts, preventing data loss, drastically lowering development time, and simplifying long-term maintenance.

Impact and Business Benefits

- **Accelerated Data Processing and Scalability:** Leveraging Databricks as a high-performance compute engine eliminated performance bottlenecks associated with traditional ETL tools, cutting pipeline execution times significantly and accommodating growing data volumes seamlessly.
- **Flawless Complex Data Extraction:** Native handling of complex array objects and semi-structured Workday API payloads ensured 100% data fidelity without custom parsing overhead or schema degradation during transformations.
- **Near-Real-Time Business Insights:** Incremental data pipelines ensured downstream platforms received fast, continuous data updates, empowering business teams to make timely, data-driven decisions based on current operational data.
- **Zero Downtime and Uninterrupted Data Access:** The staging table swap pattern eliminated table locking and query disruptions, delivering continuous,

seamless data access to end consumers and business tools during target refreshes.

- **High Pipeline Reliability and Zero Data Loss:** The metadata-driven data chunking strategy prevented API connection timeouts, eliminating data loss during bulk Workday extractions and ensuring reliable pipeline runs.
- **Reduced Maintenance and Faster Time-to-Market:** Centralizing pipeline management through a single control table streamlined development, reduced operational maintenance overhead, and made adding or modifying API feeds significantly easier.

Concluding line for CTA : Ready to transform your data management? Contact us today to learn how our MDM solutions can help your business achieve a single source of truth.